

ANCIENT NAVIER–STOKES SOLUTIONS UNDER AXISYMMETRY AND DECAY

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ABSTRACT. We consider ancient solutions of the three-dimensional Navier–Stokes equations arising in the Type I blow-up analysis under three additional structures: axisymmetry without swirl, axisymmetry with a quantitative bound on the angular velocity, and weighted spatial decay in self-similar variables. In the no-swirl case, the Liouville theorem of Koch–Nadirashvili–Seregin–Sverak, applied after a time shift, implies that every smooth Type I ancient solution is identically zero. We also use two swirl-bearing axisymmetric Liouville theorems: the Lei–Zhang–Zhao theorem for $\Gamma = rU_\theta \in L_t^\infty L_x^p$, $1 \leq p < \infty$, and the Lei–Ren–Zhang theorem for z -periodic ancient solutions with bounded Γ . The pointwise controlled-swirl and weighted alternatives are reduced to explicitly stated fixed-exponent Liouville hypotheses.

1. INTRODUCTION

Let $U : \mathbb{R}^3 \times (-\infty, 0) \rightarrow \mathbb{R}^3$ and $P : \mathbb{R}^3 \times (-\infty, 0) \rightarrow \mathbb{R}$ solve

$$(1.1) \quad \partial_t U - \Delta U + U \cdot \nabla U + \nabla P = 0, \quad \nabla \cdot U = 0.$$

The Type I analysis of a finite-time singularity leads, after rescaling about a singular point, to nonzero mild ancient solutions satisfying the Type I bound [3, 4, 8, 9]

$$(1.2) \quad \sup_{t < 0} \sqrt{-t} \|U(\cdot, t)\|_{L^\infty(\mathbb{R}^3)} < \infty.$$

Throughout the paper, a nonzero ancient limit means an ancient solution obtained from the blow-up procedure with the normalization fixed in the companion papers: the limiting solution carries a positive pressure-normalized local critical density, as in Seregin’s compactness argument [3, 4, 8]. The present paper records three additional structures under which such limits can be excluded.

The first case is axisymmetry without swirl. In this case the argument is settled by the Liouville theorem asserting that bounded ancient axisymmetric no-swirl solutions are constant. Since a solution satisfying (1.2) is bounded on every time interval $(-\infty, T]$, $T < 0$, this theorem applies after a time shift. The Type I decay as $t \rightarrow -\infty$ then eliminates the constant.

The second case is axisymmetry with swirl satisfying a quantitative bound. The available estimates for the angular component are substantially weaker than in the no-swirl case. Two cases, however, are covered by published Liouville theorems: finite L^p control of the swirl variable $\Gamma = rU_\theta$, and axial periodicity with bounded Γ . After recording these inputs, we formulate the remaining pointwise controlled-swirl criterion as a fixed-exponent Liouville hypothesis, with $\gamma > 0$ fixed throughout the statement.

The third case is uniform spatial decay in self-similar variables. This condition rules out spatial escape in weighted L^2 , but it does not, by itself, furnish a time-dependent Liouville theorem. We therefore isolate the weighted Liouville hypothesis and prove the reduction from that hypothesis.

1.1. Normalization convention. Every exclusion below is applied to an ancient limit produced by a singularity. Such a limit is a bounded ancient solution together with the nontriviality normalization inherited from the blow-up sequence: it carries a fixed positive pressure-normalized local critical density. This is the nonzero ancient-limit normalization obtained from the positive local concentration and compactness persistence statements in [3, Theorem `thm:singular-positive` and Proposition `prop:compactness-persistence`] and from the selected-scale nontriviality theorem [4, Proposition `prop:nonzero`]. Therefore any Liouville theorem which concludes $U \equiv 0$ contradicts the construction of the limit. Apart from this nontriviality normalization, the arguments below use only the stated PDE properties of the ancient solution.

Lemma 1.1 (Nonzero normalization of blow-up limits). *Let U be a Type I ancient limit obtained from the blow-up procedure used in the companion papers. Then $U \not\equiv 0$ on $\mathbb{R}^3 \times (-\infty, 0)$.*

Proof. The blow-up construction fixes a positive pressure-normalized local critical density for the limiting solution. The persistence of this positive density under the local compactness passage is [3, Proposition `prop:compactness-persistence`], and the selected Type I ancient limit is proved nonzero in [4, Proposition `prop:nonzero`]. The identically zero velocity has vanishing local critical velocity density and, under the pressure gauge convention used throughout this paper, vanishing pressure-normalized density. Hence the normalized ancient limit cannot be the zero solution. $\square$

1.2. Notation and PDE conventions. For a time interval $I \subset \mathbb{R}$ and a measurable function f , we write

$$\|f\|_{L_t^\infty L_x^p(\mathbb{R}^3 \times I)} = \operatorname{ess\,sup}_{t \in I} \|f(\cdot, t)\|_{L^p(\mathbb{R}^3)}, \quad 1 \leq p < \infty,$$

and

$$\|f\|_{L^\infty(\mathbb{R}^3 \times I)} = \operatorname{ess\,sup}_{(x,t) \in \mathbb{R}^3 \times I} |f(x, t)|.$$

All solutions considered in the proofs are smooth, so these essential suprema agree with the corresponding pointwise suprema after redefining on null sets. The subscript on the time norm records only the time variable being used; for example $L_s^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0))$ and $L_t^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0))$ denote the same mixed norm after renaming the time coordinate. The heat semigroup $e^{t\Delta}$ and the Leray projector $\mathbb{P}$ are the whole-space operators on $\mathbb{R}^3$; see, for example, Sohr's functional-analytic treatment of the Navier–Stokes equations [10]. The pressure is always understood modulo functions of time, since only ∇P appears in (1.1); invariance of the equation and the local energy inequality under such pressure gauges is recorded in [4, Lemma `lem:gauges-harmless`]. The scale-invariant pressure-normalized quantities used in the blow-up construction are those of [3, Proposition `prop:critical-scaling`]. When an axisymmetric scalar such as $\Gamma_U(r, z, t) = rU_\theta(r, z, t)$ is placed in an $L_x^p(\mathbb{R}^3)$ space, it is regarded as the corresponding function of $x = (r \cos \theta, r \sin \theta, z)$, independent of θ . Thus, for example,

$$\|\Gamma_U(\cdot, t)\|_{L^p(\mathbb{R}^3)}^p = 2\pi \int_{\mathbb{R}} \int_0^\infty |\Gamma_U(r, z, t)|^p r \, dr \, dz, \quad 1 \leq p < \infty.$$

The axis $\{r = 0\}$ has three-dimensional Lebesgue measure zero, so the value assigned to a cylindrical scalar on the axis does not affect its L^p -class; when a pointwise on-axis value is needed, it is supplied by smooth extension of the underlying Cartesian field. An axisymmetric solution is called z -periodic if there exists a number $L > 0$ such that its Cartesian velocity field satisfies

$$U(x_1, x_2, z + L, t) = U(x_1, x_2, z, t)$$

for every $(x_1, x_2, z, t) \in \mathbb{R}^3 \times (-\infty, 0)$. Equivalently, all cylindrical components are L -periodic in the axial variable z for $r > 0$; the Cartesian formulation is the definition used at the axis. No periodicity of the pressure itself is assumed; only ∇P enters the Navier-Stokes equation, and the pressure is always understood up to a time-dependent gauge.

2. ANCIENT TYPE I SOLUTIONS AND SELF-SIMILAR VARIABLES

Definition 2.1. A smooth mild solution U of (1.1) on $\mathbb{R}^3 \times (-\infty, 0)$ is called a Type I ancient solution if (1.2) holds. Here mild is understood in the standard velocity formulation, equivalently by the Duhamel formula on each interval $[a, b] \subset (-\infty, 0)$ [1, 10]:

$$(2.1) \quad U(\cdot, t) = e^{(t-s)\Delta} U(\cdot, s) - \int_s^t e^{(t-\sigma)\Delta} \mathbb{P} \nabla \cdot (U(\cdot, \sigma) \otimes U(\cdot, \sigma)) d\sigma, \quad -\infty < s < t < 0,$$

in the sense of distributions, where $\mathbb{P}$ denotes the Leray projection. Since all solutions considered below are smooth and bounded on compact subintervals of $(-\infty, 0)$, the tensor $U(\cdot, \sigma) \otimes U(\cdot, \sigma)$ is locally bounded in space-time; the Duhamel integral is therefore a well-defined distributional integral on each finite time interval. Taking the divergence of (1.1) gives the standard pressure identity

$$-\Delta P = \partial_i \partial_j (U_i U_j).$$

This pressure reconstruction and the associated time-dependent gauge ambiguity are already established for the series in [4, Lemmas `lem:gauges-harmless` and `lem:pressure`] and [5, Theorem `thm:gauges` and Lemma `lem:mild-to-projected`]; we use that convention here without reproving it. The velocity and pressure in backward self-similar variables are

$$(2.2) \quad V(y, \tau) = \sqrt{-t} U(x, t), \quad Q(y, \tau) = (-t) P(x, t), \quad y = \frac{x}{\sqrt{-t}}, \quad \tau = -\log(-t).$$

Lemma 2.2 (Self-similar equation). *Let (U, P) be a smooth solution of (1.1), and define (V, Q) by (2.2). Then V is defined on $\mathbb{R}^3 \times \mathbb{R}$ and satisfies*

$$(2.3) \quad \partial_\tau V - \Delta V + \frac{1}{2} y \cdot \nabla V + \frac{1}{2} V + V \cdot \nabla V + \nabla Q = 0, \quad \nabla \cdot V = 0.$$

Conversely, every smooth solution (V, Q) of (2.3) determines a smooth solution of (1.1) by the inverse change of variables.

Proof. The asserted equation is the backward self-similar formulation used throughout the Type I part of the series. The chain-rule computation, including the pressure scaling and the divergence equation, is given in [4, Lemma `lem:renormalized-equation`]. That lemma writes the equation as

$$\partial_\tau V + (V \cdot \nabla) V + \nabla Q = \Delta V - \frac{1}{2} (V + y \cdot \nabla V),$$

which is algebraically the same as (2.3) after moving all terms to the left-hand side. The inverse statement follows from the same diffeomorphic change of variables, also recorded there. $\square$

The Type I bound is equivalent to the uniform L^∞ bound for V :

$$(2.4) \quad \sup_{\tau \in \mathbb{R}} \|V(\cdot, \tau)\|_{L^\infty(\mathbb{R}^3)} < \infty.$$

Indeed, if $t = -e^{-\tau}$, then

$$(y, \tau) \mapsto (x, t) = (e^{-\tau/2} y, -e^{-\tau})$$

is a smooth bijection from $\mathbb{R}^3 \times \mathbb{R}$ onto $\mathbb{R}^3 \times (-\infty, 0)$, with inverse

$$y = \frac{x}{\sqrt{-t}}, \quad \tau = -\log(-t).$$

For each fixed τ , the spatial map $y \mapsto x = e^{-\tau/2}y$ is a bijection of $\mathbb{R}^3$. Therefore

$$\|V(\cdot, \tau)\|_{L^\infty(\mathbb{R}^3)} = \sqrt{-t} \|U(\cdot, t)\|_{L^\infty(\mathbb{R}^3)}.$$

Taking the supremum over $\tau \in \mathbb{R}$ is therefore the same as taking the supremum over $t < 0$, proving the equivalence of (1.2) and (2.4).

For axisymmetric flows we write

$$x = (r \cos \theta, r \sin \theta, z), \quad U = U_r e_r + U_\theta e_\theta + U_z e_z.$$

These component functions are defined on the open set $r > 0$. Since the velocity is smooth on $\mathbb{R}^3$, every identity involving cylindrical components is first interpreted for $r > 0$ and then extended to the axis by smoothness when an on-axis value is needed. The flow is said to have no swirl if $U_\theta \equiv 0$ for $r > 0$. The same convention is used for V , with cylindrical coordinates

$$y = (\rho \cos \theta, \rho \sin \theta, Z).$$

We write

$$\Gamma_U(r, z, t) = r U_\theta(r, z, t), \quad \Gamma_V(\rho, Z, \tau) = \rho V_\theta(\rho, Z, \tau).$$

Under the self-similar change of variables these quantities agree: $\Gamma_V(\rho, Z, \tau) = \Gamma_U(r, z, t)$, where $(r, z) = \sqrt{-t}(\rho, Z)$.

Lemma 2.3 (Preservation of the structured hypotheses). *Axisymmetry and the no-swirl condition are preserved by the self-similar change of variables (2.2). For any fixed $\gamma > 0$, the bounds*

$$(2.5) \quad \sup_{\tau \in \mathbb{R}} \sup_{\rho > 0, Z \in \mathbb{R}} \rho^\gamma |V_\theta(\rho, Z, \tau)| < \infty$$

are invariant under translations of τ . If U is axisymmetric, then the self-similar swirl variables satisfy

$$\Gamma_V(\rho, Z, \tau) = \Gamma_U(r, z, t), \quad (r, z) = \sqrt{-t}(\rho, Z).$$

Moreover, if U satisfies (1.2), then for every $T < 0$ the restriction of U to $\mathbb{R}^3 \times (-\infty, T]$ is bounded.

Proof. The self-similar change of variables is a scalar dilation in space together with a scalar multiplication of the velocity. It therefore commutes with rotations around the z -axis. More explicitly, for every rotation R_α around the symmetry axis,

$$U(R_\alpha x, t) = R_\alpha U(x, t)$$

implies

$$V(R_\alpha y, \tau) = \sqrt{-t} U(R_\alpha x, t) = R_\alpha \sqrt{-t} U(x, t) = R_\alpha V(y, \tau).$$

Thus axisymmetry is preserved. Since the cylindrical basis (e_r, e_θ, e_z) is unchanged by a radial dilation, the component relation is

$$V_\theta(\rho, Z, \tau) = \sqrt{-t} U_\theta(r, z, t), \quad r = \sqrt{-t}\rho, \quad z = \sqrt{-t}Z.$$

Consequently $U_\theta \equiv 0$ if and only if $V_\theta \equiv 0$, and

$$\Gamma_V(\rho, Z, \tau) = \rho V_\theta(\rho, Z, \tau) = \rho \sqrt{-t} U_\theta(r, z, t) = r U_\theta(r, z, t) = \Gamma_U(r, z, t).$$

The expression $\rho^\gamma |V_\theta|$ is written directly in self-similar coordinates, so (2.5) is unchanged when $V(y, \tau)$ is replaced by $V(y, \tau + a)$, $a \in \mathbb{R}$; the supremum is over all $\tau \in \mathbb{R}$. Finally, if $t \leq T < 0$, then (1.2) gives

$$\|U(\cdot, t)\|_{L^\infty(\mathbb{R}^3)} \leq \frac{1}{\sqrt{-T}} \sup_{s < 0} \sqrt{-s} \|U(\cdot, s)\|_{L^\infty(\mathbb{R}^3)}.$$

□

Lemma 2.4 (Time shifts away from the singular time). *Let U be a smooth Type I ancient solution of (1.1). Fix $T < 0$, and set*

$$W_T(x, s) = U(x, T + s), \quad \Pi_T(x, s) = P(x, T + s), \quad s < 0.$$

Then (W_T, Π_T) is a smooth bounded mild ancient solution of (1.1) on $\mathbb{R}^3 \times (-\infty, 0)$. If U is axisymmetric, has no swirl, is periodic in z , or satisfies an $L_t^\infty L_x^p$ or L^∞ bound on $\Gamma_U = rU_\theta$, then W_T has the same corresponding property on $\mathbb{R}^3 \times (-\infty, 0)$.

Proof. Smoothness and the differential equations are invariant under translations in time, since $\partial_s W_T(x, s) = \partial_t U(x, T + s)$, all spatial derivatives are unchanged, and

$$\nabla \Pi_T(x, s) = \nabla P(x, T + s).$$

Consequently

$$\partial_s W_T - \Delta W_T + W_T \cdot \nabla W_T + \nabla \Pi_T = (\partial_t U - \Delta U + U \cdot \nabla U + \nabla P)(x, T + s) = 0,$$

and $\nabla \cdot W_T = (\nabla \cdot U)(x, T + s) = 0$. The mild identity follows by applying (2.1) to U at the times $T + s < T + t < 0$. Namely, for $-\infty < s < t < 0$,

$$U(\cdot, T + t) = e^{(t-s)\Delta} U(\cdot, T + s) - \int_{T+s}^{T+t} e^{(T+t-\eta)\Delta} \mathbb{P} \nabla \cdot (U(\cdot, \eta) \otimes U(\cdot, \eta)) d\eta.$$

Set $\eta = T + \sigma$. Then $d\eta = d\sigma$, the interval $[T + s, T + t]$ becomes $[s, t]$, and $T + t - \eta = t - \sigma$. Since $U(\cdot, T + \sigma) = W_T(\cdot, \sigma)$, the preceding identity becomes

$$W_T(\cdot, t) = e^{(t-s)\Delta} W_T(\cdot, s) - \int_s^t e^{(t-\sigma)\Delta} \mathbb{P} \nabla \cdot (W_T(\cdot, \sigma) \otimes W_T(\cdot, \sigma)) d\sigma,$$

which is the mild formulation for W_T . Boundedness follows from Theorem 2.3:

$$\|W_T(\cdot, s)\|_{L^\infty(\mathbb{R}^3)} = \|U(\cdot, T + s)\|_{L^\infty(\mathbb{R}^3)} \leq \frac{1}{\sqrt{-T}} \sup_{\eta < 0} \sqrt{-\eta} \|U(\cdot, \eta)\|_{L^\infty(\mathbb{R}^3)}.$$

Axisymmetry is unchanged because for every rotation R_α around the z -axis,

$$W_T(R_\alpha x, s) = U(R_\alpha x, T + s) = R_\alpha U(x, T + s) = R_\alpha W_T(x, s).$$

Absence of swirl is unchanged because

$$(W_T)_\theta(r, z, s) = U_\theta(r, z, T + s),$$

so $U_\theta \equiv 0$ for $r > 0$ implies $(W_T)_\theta \equiv 0$ for $r > 0$. If U is L -periodic in z , then

$$W_T(x_1, x_2, z + L, s) = U(x_1, x_2, z + L, T + s) = U(x_1, x_2, z, T + s) = W_T(x_1, x_2, z, s),$$

so the same period works for W_T . Finally

$$\Gamma_{W_T}(r, z, s) = r(W_T)_\theta(r, z, s) = rU_\theta(r, z, T + s) = \Gamma_U(r, z, T + s).$$

Therefore an $L_t^\infty L_x^p$ bound for Γ_U on $(-\infty, 0)$ restricts to the shifted solution, since

$$\|\Gamma_{W_T}\|_{L_s^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0))} = \operatorname{ess\,sup}_{s < 0} \|\Gamma_U(\cdot, T + s)\|_{L^p(\mathbb{R}^3)} \leq \|\Gamma_U\|_{L_t^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0))}.$$

Similarly,

$$\|\Gamma_{W_T}\|_{L^\infty(\mathbb{R}^3 \times (-\infty, 0))} \leq \|\Gamma_U\|_{L^\infty(\mathbb{R}^3 \times (-\infty, 0))}.$$

□

Lemma 2.5 (Spatially constant mild solutions). *Let W be a smooth mild solution of (1.1) on $\mathbb{R}^3 \times (-\infty, 0)$. If there is a function $b : (-\infty, 0) \rightarrow \mathbb{R}^3$ such that $W(x, t) = b(t)$ for all (x, t) , then b is constant in time.*

Proof. Fix $-\infty < s < t < 0$. Applying the mild formula (2.1) to W gives

$$b(t) = e^{(t-s)\Delta}b(s) - \int_s^t e^{(t-\sigma)\Delta} \mathbb{P} \nabla \cdot (b(\sigma) \otimes b(\sigma)) d\sigma.$$

The heat semigroup leaves spatially constant vector fields fixed. Moreover $b(\sigma) \otimes b(\sigma)$ is a spatially constant matrix, hence its distributional divergence is zero. The Duhamel integral therefore vanishes, and $b(t) = b(s)$. Since $s < t$ were arbitrary, b is constant on $(-\infty, 0)$. $\square$

Lemma 2.6 (Patching constants on backward time intervals). *Let $U : \mathbb{R}^3 \times (-\infty, 0) \rightarrow \mathbb{R}^3$ be any function. Assume that for every $T < 0$ there exists a vector $c_T \in \mathbb{R}^3$ such that*

$$U(x, t) = c_T \quad \text{for all } x \in \mathbb{R}^3, \quad t < T.$$

Then there is a single vector $c \in \mathbb{R}^3$ such that

$$U(x, t) = c \quad \text{for all } x \in \mathbb{R}^3, \quad t < 0.$$

Proof. Let $T_1 < T_2 < 0$. The two representations hold simultaneously on the nonempty time interval $(-\infty, T_1)$. Hence, for any fixed $x_0 \in \mathbb{R}^3$ and any $t_0 < T_1$,

$$c_{T_1} = U(x_0, t_0) = c_{T_2}.$$

Thus all constants c_T are equal; denote their common value by c . If $t < 0$, choose T with $t < T < 0$. The hypothesis for this T gives $U(x, t) = c_T = c$ for every $x \in \mathbb{R}^3$. $\square$

Lemma 2.7 (Type I constants vanish). *Let U be a smooth Type I ancient solution of (1.1). If there is a vector $c \in \mathbb{R}^3$ such that $U(x, t) = c$ on $\mathbb{R}^3 \times (-\infty, 0)$, then $c = 0$.*

Proof. Let

$$M = \sup_{t < 0} \sqrt{-t} \|U(\cdot, t)\|_{L^\infty(\mathbb{R}^3)} < \infty.$$

For a constant solution, $\|U(\cdot, t)\|_{L^\infty} = |c|$ for every $t < 0$. Hence

$$\sqrt{-t} |c| \leq M \quad \text{for every } t < 0.$$

If $c \neq 0$, the left-hand side tends to $+\infty$ as $t \rightarrow -\infty$, contradicting the finiteness of M . Thus $c = 0$. $\square$

3. THE AXISYMMETRIC NO-SWIRL CASE

We use the following classical Liouville theorem in the precise form needed below.

Theorem 3.1 (Koch–Nadirashvili–Seregin–Sverak). *Let W be a bounded smooth mild ancient solution of the three-dimensional Navier–Stokes equations on $\mathbb{R}^3 \times (-\infty, 0)$. If W is axisymmetric without swirl, then W is constant in space and time.*

Proof. This is precisely the axisymmetric no-swirl Liouville theorem proved in Section 5 of [2]. Their hypothesis is stated for bounded ancient weak solutions. A smooth bounded mild solution satisfies the Navier–Stokes equations distributionally and therefore is admissible for that theorem; no additional compactness or decay assumption is used here. If the conclusion is expressed as a spatially constant axial drift, the mild formulation makes the drift independent of time by Theorem 2.5. $\square$

Proposition 3.2. *Let U be a smooth Type I ancient solution of (1.1). If U is axisymmetric without swirl, then $U \equiv 0$.*

Proof. Fix $T < 0$, and define

$$W_T(x, s) = U(x, T + s), \quad s < 0.$$

By [Theorem 2.4](#), W_T is a bounded smooth mild ancient solution on $\mathbb{R}^3 \times (-\infty, 0)$, and it is axisymmetric without swirl. By [Theorem 3.1](#), W_T is constant on $\mathbb{R}^3 \times (-\infty, 0)$; write this constant as c_T . Equivalently,

$$U(x, t) = c_T \quad \text{for } x \in \mathbb{R}^3, \quad t < T.$$

Since $T < 0$ was arbitrary, [Theorem 2.6](#) gives a single vector $c \in \mathbb{R}^3$ such that $U(x, t) = c$ for every $x \in \mathbb{R}^3$ and every $t < 0$. By [Theorem 2.7](#), $c = 0$. $\square$

Corollary 3.3 (Unconditional no-swirl Type I exclusion). *No nonzero Type I ancient limit obtained by the Seregin blow-up procedure from a finite-time Type I singularity can be axisymmetric without swirl.*

Proof. By the definition used in the blow-up construction, the ancient limit is a smooth Type I ancient solution and is not identically zero: the normalization fixes a positive pressure-normalized local critical density, as recalled in the normalization convention above and proved in [[3](#), [Theorem thm:singular-positive](#) and [Proposition prop:compactness-persistence](#)] and [[4](#), [Proposition prop:nonzero](#), [Lemma lem:smoothness](#), and [Corollary cor:classical-type-I-bound](#)]. If this limit were axisymmetric without swirl, [Theorem 3.2](#) would imply $U \equiv 0$. This contradicts the nonzero normalization in [Theorem 1.1](#), and hence the no-swirl alternative cannot occur. $\square$

4. AXISYMMETRIC SOLUTIONS WITH CONTROLLED SWIRL

The angular component satisfies an equation with singular coefficients on the axis. Thus the no-swirl Liouville argument does not by itself treat solutions with $U_\theta \not\equiv 0$. We first record two unconditional Liouville inputs from the literature.

Theorem 4.1 (Lei–Zhang–Zhao). *Let W be a bounded smooth mild ancient solution of [\(1.1\)](#) on $\mathbb{R}^3 \times (-\infty, 0)$. Assume that W is axisymmetric and that, for some $1 \leq p < \infty$,*

$$rW_\theta \in L_t^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0)).$$

Then W is constant in space and time.

Proof. The stated result is the axisymmetric ancient Liouville theorem with swirl of Lei–Zhang–Zhao [[7](#)]. In the form used here, the theorem assumes that W is a bounded smooth ancient mild solution and that $\Gamma = rW_\theta \in L_t^\infty L_x^p$ for some $1 \leq p < \infty$; its conclusion is that W is spatially constant. The mild formulation then upgrades spatial constancy to constancy in time by [Theorem 2.5](#). $\square$

Proposition 4.2 (Type I exclusion under L^p -controlled swirl). *Let U be a smooth Type I ancient solution of [\(1.1\)](#). If U is axisymmetric and, for some $1 \leq p < \infty$,*

$$\Gamma_U = rU_\theta \in L_t^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0)),$$

then $U \equiv 0$.

Proof. Fix $T < 0$ and set $W_T(x, s) = U(x, T + s)$, $s < 0$. By [Theorem 2.4](#), W_T is a bounded smooth mild ancient solution on $\mathbb{R}^3 \times (-\infty, 0)$, it is axisymmetric, and

$$r(W_T)_\theta \in L_s^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0)).$$

The hypotheses of [Theorem 4.1](#) are therefore satisfied by W_T , and W_T is a constant vector field. Thus $U(x, t) = c_T$ on $\mathbb{R}^3 \times (-\infty, T)$. Since $T < 0$ was arbitrary, [Theorem 2.6](#) gives a single vector c with $U(x, t) = c$ for all $x \in \mathbb{R}^3$ and all $t < 0$. [Theorem 2.7](#) then gives $c = 0$. $\square$

Theorem 4.3 (Lei–Ren–Zhang). *Let W be a bounded smooth mild ancient solution of (1.1) on $\mathbb{R}^3 \times (-\infty, 0)$. Assume that W is axisymmetric, periodic in the axial variable z , and*

$$rW_\theta \in L^\infty(\mathbb{R}^3 \times (-\infty, 0)).$$

Then W is constant in space and time.

Proof. We use the periodic axisymmetric Liouville theorem of Lei–Ren–Zhang [6]. Their theorem is stated on $\mathbb{R}^2 \times \mathbb{T}^1$; a solution on $\mathbb{R}^3$ which is periodic in z descends to the corresponding quotient as follows. If $L > 0$ is an axial period of the velocity, then

$$W(x_1, x_2, z + L, t) = W(x_1, x_2, z, t)$$

for every (x_1, x_2, z, t) . Hence the formula

$$\widetilde{W}(x_1, x_2, [z], t) = W(x_1, x_2, z, t), \quad [z] \in \mathbb{R}/L\mathbb{Z},$$

is independent of the representative z and defines a smooth vector field on $\mathbb{R}^2 \times (\mathbb{R}/L\mathbb{Z})$. Smoothness on the quotient is local in the axial coordinate and follows from the smoothness and periodicity of the Cartesian field W ; no separate cylindrical-coordinate argument at the axis is required. The pressure gradient also descends to the quotient. Indeed, from the smooth equation

$$\nabla \Pi = -\partial_t W + \Delta W - W \cdot \nabla W$$

the right-hand side is L -periodic in z : spatial derivatives preserve periodicity, and differentiating the identity $W(x_1, x_2, z + L, t) = W(x_1, x_2, z, t)$ in t gives periodicity of $\partial_t W$. Hence $\nabla \Pi$ is a well-defined one-form on the quotient. Since the Navier–Stokes system is a local differential system in the axial variable, the descended velocity together with this descended pressure gradient solves the same system on the quotient. This use of the pressure is consistent with the pressure-gauge convention above: the theorem is applied through the velocity equation involving $\nabla \Pi$. To match the period of the torus used in [6], namely 2π , use the Navier–Stokes scaling

$$W_\lambda(x, t) = \lambda W(\lambda x, \lambda^2 t), \quad \Pi_\lambda(x, t) = \lambda^2 \Pi(\lambda x, \lambda^2 t),$$

with $\lambda = L/(2\pi)$. Then

$$W_\lambda(x_1, x_2, z + 2\pi, t) = \lambda W(\lambda x_1, \lambda x_2, \lambda z + L, \lambda^2 t) = W_\lambda(x_1, x_2, z, t),$$

so the scaled solution is defined on $\mathbb{R}^2 \times (\mathbb{R}/2\pi\mathbb{Z}) = \mathbb{R}^2 \times \mathbb{T}^1$. The fact that this parabolic scaling preserves the Navier–Stokes equation, the divergence condition, and pressure gauges is the scaling statement proved earlier in [4, Lemmas `lem:scaling` and `lem:gauges-harmless`], together with the scale-invariance of the pressure-normalized local quantities in [3, Proposition `prop:critical-scaling`]; we use it here only to normalize the axial period. Axisymmetry is preserved because scalar dilations commute with rotations around the z -axis. Boundedness is preserved because

$$\|W_\lambda\|_{L^\infty(\mathbb{R}^3 \times (-\infty, 0))} = \lambda \|W\|_{L^\infty(\mathbb{R}^3 \times (-\infty, 0))} < \infty.$$

Ancientness is preserved because $t < 0$ if and only if $\lambda^2 t < 0$. The mild formulation is preserved by applying the same change of variables to the Duhamel formula. Equivalently, the scaled pair is a smooth solution of the differential Navier–Stokes system on every finite time interval and therefore has the standard mild representation on that interval [1, 10]. Finally, the scaling preserves boundedness of the swirl variable because

$$r(W_\lambda)_\theta(r, z, t) = r\lambda W_\theta(\lambda r, \lambda z, \lambda^2 t) = (\lambda r)W_\theta(\lambda r, \lambda z, \lambda^2 t).$$

All hypotheses of the Lei–Ren–Zhang theorem are therefore satisfied on $\mathbb{R}^2 \times \mathbb{T}^1$. Their conclusion gives an axial constant field, or equivalently a spatially constant drift. By [Theorem 2.5](#) this drift is independent of time. Thus, if $W_\lambda \equiv c_\lambda$, then for every x and t ,

$$W(x, t) = \lambda^{-1} W_\lambda(\lambda^{-1} x, \lambda^{-2} t) = \lambda^{-1} c_\lambda.$$

Thus the original lifted solution W is also constant on $\mathbb{R}^3 \times (-\infty, 0)$. $\square$

Proposition 4.4 (Type I exclusion under periodic bounded swirl). *Let U be a smooth Type I ancient solution of [\(1.1\)](#). If U is axisymmetric, periodic in z , and*

$$\Gamma_U = rU_\theta \in L^\infty(\mathbb{R}^3 \times (-\infty, 0)),$$

then $U \equiv 0$.

Proof. Fix $T < 0$ and set $W_T(x, s) = U(x, T + s)$, $s < 0$. By [Theorem 2.4](#), W_T is a bounded smooth mild ancient solution, axisymmetric, periodic in z , and with

$$r(W_T)_\theta \in L^\infty(\mathbb{R}^3 \times (-\infty, 0)).$$

Applying [Theorem 4.3](#) gives that W_T is constant on $\mathbb{R}^3 \times (-\infty, 0)$; write the corresponding constant as c_T . Therefore

$$U(x, t) = c_T \quad \text{for } x \in \mathbb{R}^3, \quad t < T.$$

Since $T < 0$ was arbitrary, [Theorem 2.6](#) yields a global constant vector c on $\mathbb{R}^3 \times (-\infty, 0)$. By [Theorem 2.7](#), $c = 0$. $\square$

The remaining controlled-swirl case is not covered by the preceding imported Liouville theorems. We isolate the exact fixed-exponent Liouville input needed for this case.

Hypothesis 4.5 (Liouville theorem with controlled swirl). Fix $\gamma > 0$. Let U be a smooth Type I ancient solution of [\(1.1\)](#). Assume that U is axisymmetric and that its velocity V in self-similar variables satisfies

$$(4.1) \quad \sup_{\tau \in \mathbb{R}} \sup_{\rho > 0, Z \in \mathbb{R}} \rho^\gamma |V_\theta(\rho, Z, \tau)| < \infty.$$

Then $U \equiv 0$.

Remark 4.6. The exponent γ is part of [Theorem 4.5](#). The result is conditional at this value of γ and makes no assertion at other exponents.

Proposition 4.7. *Fix $\gamma > 0$, and suppose that the Liouville hypothesis [Theorem 4.5](#) holds for this value of γ . Let U be a nonzero Type I ancient limit arising from a finite-time Type I singularity. Then U cannot be axisymmetric with self-similar velocity satisfying [\(4.1\)](#).*

Proof. The parameter in the proposition is the same γ as in [Theorem 4.5](#). By the blow-up construction cited in the normalization convention, specifically [\[4, Proposition prop:nonzero, Lemma lem:smoothness, and Corollary cor:classical-type-I-bound\]](#), U is a smooth Type I ancient solution and is not identically zero. The additional hypotheses in the proposition are precisely axisymmetry and [\(4.1\)](#). Thus all hypotheses of [Theorem 4.5](#) are satisfied with the same exponent γ , and no decay, periodicity, or integrability condition beyond those stated in [Theorem 4.5](#) is used. Hence [Theorem 4.5](#) gives $U \equiv 0$. This contradicts the nonzero normalization of the Type I ancient limit in [Theorem 1.1](#). $\square$

Remark 4.8. In physical variables, (4.1) is equivalent to

$$\sup_{t < 0} \sup_{r > 0, z \in \mathbb{R}} (-t)^{(1-\gamma)/2} r^\gamma |U_\theta(r, z, t)| < \infty.$$

Indeed, $r = \sqrt{-t}\rho$ and $V_\theta(\rho, Z, \tau) = \sqrt{-t}U_\theta(r, z, t)$. Therefore

$$\rho^\gamma |V_\theta(\rho, Z, \tau)| = \left(\frac{r}{\sqrt{-t}} \right)^\gamma \sqrt{-t} |U_\theta(r, z, t)| = (-t)^{(1-\gamma)/2} r^\gamma |U_\theta(r, z, t)|.$$

The change of variables $(r, z, t) \leftrightarrow (\rho, Z, \tau)$ is bijective between $\{r > 0, z \in \mathbb{R}, t < 0\}$ and $\{\rho > 0, Z \in \mathbb{R}, \tau \in \mathbb{R}\}$, so the two suprema are equal. Thus (4.1) is equivalent to the displayed scale-invariant physical-variable bound. We state [Theorem 4.5](#) in self-similar variables so that the geometric cylindrical weight and the Type I time weight are separated.

5. FAST DECAY IN SELF-SIMILAR VARIABLES

We next formulate the spatial decay case. Let V be the velocity in self-similar variables associated with a Type I ancient solution. For $m > 0$ consider the weighted condition

$$(5.1) \quad \sup_{\tau \in \mathbb{R}} \int_{\mathbb{R}^3} (1 + |y|^2)^m |V(y, \tau)|^2 dy < \infty.$$

Let

$$A_m = \sup_{\tau \in \mathbb{R}} \int_{\mathbb{R}^3} (1 + |y|^2)^m |V(y, \tau)|^2 dy.$$

Then (5.1) implies uniform local L^2 control. Indeed, since $m > 0$, the weight satisfies $(1 + |y|^2)^m \geq 1$ for every y . Hence, for every $R < \infty$ and every τ ,

$$\int_{B_R} |V(y, \tau)|^2 dy \leq \int_{\mathbb{R}^3} (1 + |y|^2)^m |V(y, \tau)|^2 dy.$$

Taking the supremum over τ gives

$$\sup_{\tau \in \mathbb{R}} \int_{B_R} |V(y, \tau)|^2 dy \leq A_m.$$

It also rules out spatial escape of L^2 mass in self-similar variables: for every $R > 0$, the implication $|y| > R$ gives $(1 + |y|^2)^m \geq (1 + R^2)^m$. Therefore, for every τ ,

$$\int_{|y| > R} |V(y, \tau)|^2 dy \leq (1 + R^2)^{-m} \int_{\mathbb{R}^3} (1 + |y|^2)^m |V(y, \tau)|^2 dy.$$

Taking the supremum over τ gives

$$\sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^2 dy \leq (1 + R^2)^{-m} A_m.$$

Thus the right-hand side tends to zero as $R \rightarrow \infty$. The present paper does not use any integration by parts or backward-uniqueness argument beyond the fixed-weight Liouville input stated next.

Hypothesis 5.1 (Weighted Liouville theorem). Fix $m > 0$. Let U be a smooth Type I ancient solution of (1.1). Assume that its velocity V in self-similar variables satisfies (5.1). Then $U \equiv 0$.

Remark 5.2. The weight m is part of [Theorem 5.1](#). The result is conditional at this value of m and makes no assertion at other weights.

Proposition 5.3. *Fix $m > 0$, and suppose that the Liouville hypothesis [Theorem 5.1](#) holds for this value of m . Let U be a nonzero Type I ancient limit arising from a finite-time Type I singularity. Then its velocity in self-similar variables cannot satisfy [\(5.1\)](#).*

Proof. The parameter in the proposition is the same m as in [Theorem 5.1](#). By the blow-up construction cited in the normalization convention, specifically [\[4, Proposition `prop:nonzero`, Lemma `lem:smoothness`, and Corollary `cor:classical-type-I-bound`\]](#), U is a smooth Type I ancient solution and is not identically zero. If its self-similar velocity V satisfied [\(5.1\)](#), then all hypotheses of [Theorem 5.1](#) would be satisfied with the same weight m . Therefore $U \equiv 0$, contradicting the nonzero normalization inherited from the singular blow-up sequence, as stated in [Theorem 1.1](#). $\square$

6. THE TYPE I EXCLUSION UNDER SYMMETRY AND DECAY

The preceding sections give the following exclusion statement. The no-swirl alternative, the L^p -controlled swirl alternative, and the periodic bounded- Γ alternative are consequences of published Liouville theorems. The pointwise controlled-swirl and decay alternatives are conditional on the corresponding fixed-exponent Liouville hypotheses stated above.

Theorem 6.1. *Let U be a nonzero Type I ancient limit arising from a finite-time Type I singularity of the three-dimensional Navier–Stokes equations, and let V be its velocity in self-similar variables. Then U is not axisymmetric without swirl. Moreover, if U is axisymmetric, then neither of the following additional alternatives can occur:*

$$rU_\theta \in L_t^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0)) \quad \text{for some } 1 \leq p < \infty,$$

or U is periodic in z and

$$rU_\theta \in L^\infty(\mathbb{R}^3 \times (-\infty, 0)).$$

Finally, fix $\gamma > 0$ and suppose that the Liouville hypothesis in [Theorem 4.5](#) holds. Then U cannot be axisymmetric and satisfy [\(4.1\)](#). Likewise, fix $m > 0$ and suppose that the Liouville hypothesis in [Theorem 5.1](#) holds. Then V cannot satisfy [\(5.1\)](#).

Proof. Suppose first that U were axisymmetric without swirl. Since U is a nonzero Type I ancient limit obtained from the blow-up procedure, it is in the class covered by [Theorem 3.3](#). That corollary excludes precisely such limits under the no-swirl hypothesis. This gives a contradiction, and hence the no-swirl alternative cannot occur.

Assume next that U is axisymmetric and that $\Gamma_U = rU_\theta \in L_t^\infty L_x^p$ for some $1 \leq p < \infty$. The preceding paragraph and the companion-paper normalization give the PDE properties needed to use [Theorem 4.2](#): U is a smooth Type I ancient solution, it is axisymmetric, and the displayed $L_t^\infty L_x^p$ -bound is exactly the integrability hypothesis in that proposition. Hence [Theorem 4.2](#) gives $U \equiv 0$. This contradicts the nonzero normalization in [Theorem 1.1](#).

If U is periodic in z and $\Gamma_U \in L^\infty$, then the hypotheses of [Theorem 4.4](#) are likewise exactly the smooth Type I ancient property, axisymmetry, axial periodicity, and boundedness of rU_θ . Therefore [Theorem 4.4](#) gives $U \equiv 0$, again contradicting [Theorem 1.1](#).

Now fix $\gamma > 0$ and assume [Theorem 4.5](#). If the displayed pointwise self-similar swirl bound held, then U would be a nonzero Type I ancient limit which is axisymmetric and satisfies [\(4.1\)](#) with the same exponent γ . These are precisely the hypotheses of [Theorem 4.7](#), including the fixed value of γ . That proposition says that no such limit exists, a contradiction.

Finally, fix $m > 0$ and assume [Theorem 5.1](#). If [\(5.1\)](#) held, then U would be a nonzero Type I ancient limit whose self-similar velocity satisfies the weighted condition with the same exponent m . These are precisely the hypotheses of [Theorem 5.3](#). That proposition excludes such a

limit, again giving a contradiction. These contradictions prove all alternatives stated in the theorem. $\square$

Remark 6.2. The no-swirl case is an unconditional exclusion in the Type I blow-up analysis. The $L_t^\infty L_x^p$ -controlled swirl case and the z -periodic bounded- Γ case are also excluded by existing Liouville theorems. In the pointwise controlled-swirl and weighted cases, the only additional input is the corresponding fixed-exponent Liouville theorem.

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